🚚 Vehicle Routing Optimization (CVRP) App

1. Background
Efficient delivery and logistics are crucial in supply chain management. Companies often face challenges in planning routes for multiple vehicles to serve a set of customers while minimizing travel distance and respecting vehicle capacity constraints. The **Capacitated Vehicle Routing Problem (CVRP)** is a classic optimization problem that models this scenario.

This app allows users to **optimize delivery routes** using OR-Tools, visualize them on a map, and download the results for practical use.

2. Problem Statement
Given a set of customer locations with demands and a fleet of vehicles with fixed capacities:

- Determine the best routes for each vehicle
- Minimize the total travel distance
- Ensure that no vehicle exceeds its capacity
- Return to the depot at the end of each route

Goal: Reduce transportation costs and improve operational efficiency.

3. Features

- Load customer location data from CSV or use the default dataset
- Define the number of vehicles and vehicle capacity
- Solve the CVRP using Google OR-Tools
- Display **summary KPIs**:
  - Total vehicles used
  - Maximum load per vehicle
  - Total distance traveled
- Visualize optimized routes on a **map** (PyDeck)
- Download optimized routes as **CSV** or **JSON**
- Sidebar information about the developer and contact links

4. Input
The app accepts a CSV file with the following columns:

Column Name	Description
id	Unique location ID
lat	Latitude of location
lon	Longitude of location
demand	Demand at the location

Default dataset: `routes.csv` is provided for quick testing.

5. Output

- **KPI summary cards** showing total vehicles, max load, and total distance
- **Route list** showing each vehicle's path, load, and distance
- **Map visualization** of routes
- **Downloadable files**:
  - optimized_routes.csv
  - optimized_routes.json

6. How it Works

1. The app loads customer data (default or uploaded CSV)
2. It computes a **distance matrix** using Euclidean distance
3. Google **OR-Tools** solves the CVRP:
   - Assigns routes to each vehicle
   - Respects vehicle capacity
   - Minimizes total distance
4. Routes are displayed, KPIs are calculated, and a map visualization is generated
5. Users can download the results for further analysis

7. Dependencies

- streamlit
- pandas
- numpy
- pydeck
- ortools

8. Author

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